

Gabriel Dougherty

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Work Experience

Senior Software Engineer – FX Orders

September 2025 – Present

Bloomberg LP, New York, NY

- Working on the FXGO Orders platform on the Bloomberg Terminal in C++.

Software Engineer – Tech Lead, Configuration Tooling

April 2022 – August 2025

FIS (formerly Torstone Technology), New York, NY

- Lead implementation of a Docker, Python and Poetry-based configuration tooling system used by all developers on the Post Trade Processing (PTP) product.
- Adapt the configuration tooling system to work under the specific constraints of a production rootless container environment.
- Create a C# ASP .NET Core REST API service to import client address model data, wrapping a legacy blocking queue API using C# asynchronous primitives.
- Design and implement a complete C++ integration to send daily investor communications on 5 million positions to 1.5 million retail brokerage customers.
- Coordinate with UI and Settlement teams when implementing REST APIs for Middle Office block trades and Asset Servicing data.
- Develop enhancements to a core C++ platform code generator, written in Python, that generates the REST model serialization layer so developers do not need to handwrite it.

Lead Software Engineer – Promoted from SWE II & SWE I

May 2019– April 2022

Cadence Design Systems, Pittsburgh, PA

- Built a C++/Python machine learning and optimization framework and helped integrate it into Cadence products for analog integrated circuit (IC) design, microwave design, and fluid dynamics simulation.
- Reduced simulation time by 80% for a specific customer mixed signals design problem by integrating and tuning a genetic algorithm.
- Created REST APIs to power analytics tools for large IC design teams to collaborate on and explore optimization progress on complex designs.
- Created tools to detect and prevent overlapping regions in large, deeply nested analog IC designs using an efficient minimum bounding box algorithm.

Software Engineering Intern

May 2018 – August 2018

Acutec Precision Aerospace, Meadville, PA

- Developed a C# application to automate documentation for every finished part produced.
- Reduced documentation generation time by 90% in contrast to manual documentation.

Skills

- **Languages:** Python, C++, C#, SQL, Bash
- **Software:** Docker, Postgres/Oracle SQL, multithreading, TCP/IP, Git, Linux, CMake

Education

Edinboro University of Pennsylvania

- B.S., Computer Science

Graduation: May 2019